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ARCHITECTURE PLAN · REV 1 · 10 OCTOBER 2026

Agentic Media Platform

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THE PLAN IN ONE LINE

Orchestrate models, ground the claims, then pay for media.

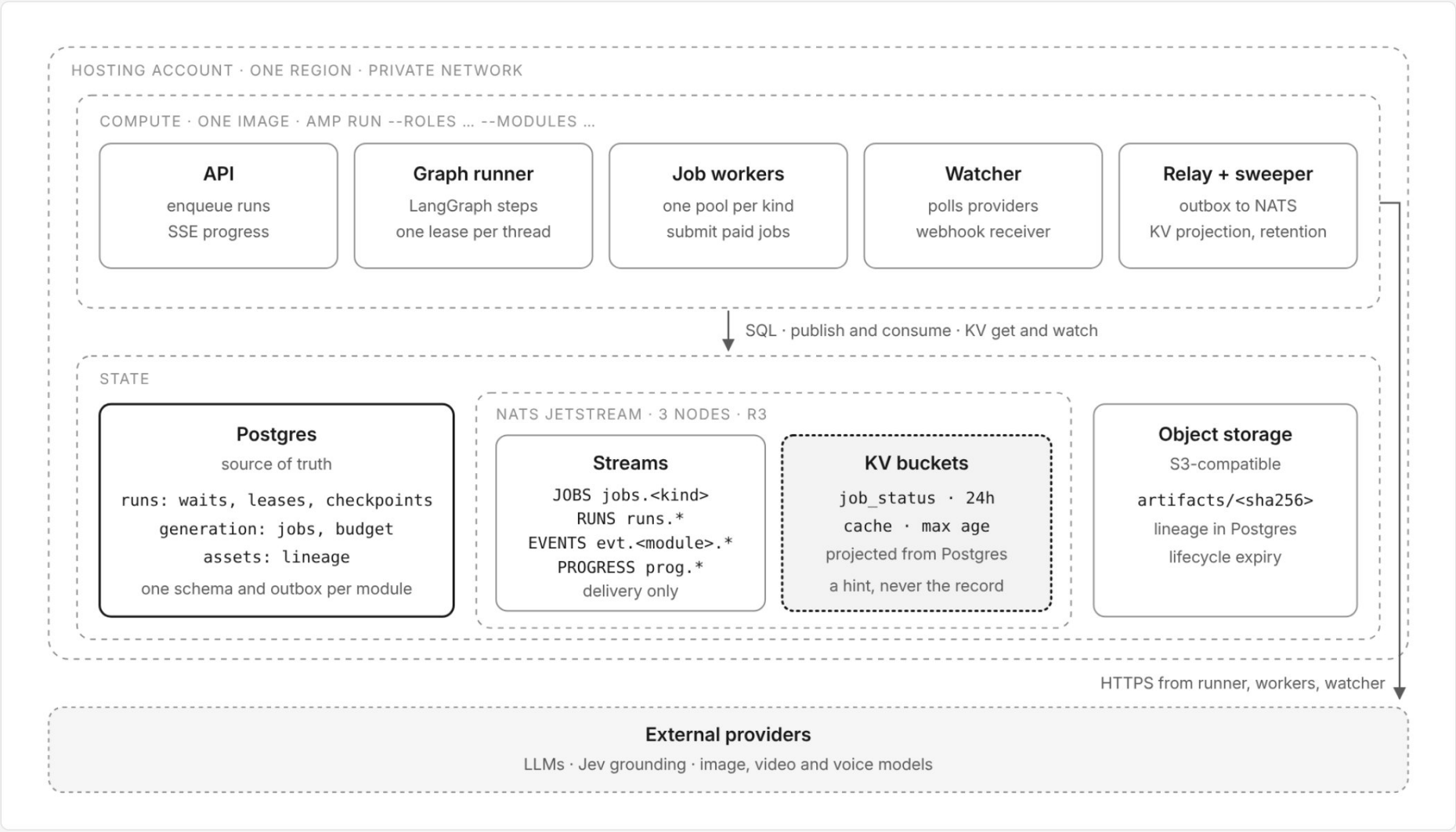
LangGraph runs multi-step workflows across several LLM providers.

Jev checks each factual claim against its evidence.

Images, video and voice arrive as long-running jobs.

Everything runs in a single region, with Postgres as the record

NATS delivers and its KV view is only a hint, so losing either is recoverable from Postgres.



Eight principles decide every design question

01

Postgres is the record; NATS delivers

02

Thin nodes, injected dependencies

03

References, not bytes

04

Every paid call is idempotent

05

Cheap gates before expensive steps

06

Modules now, services later

07

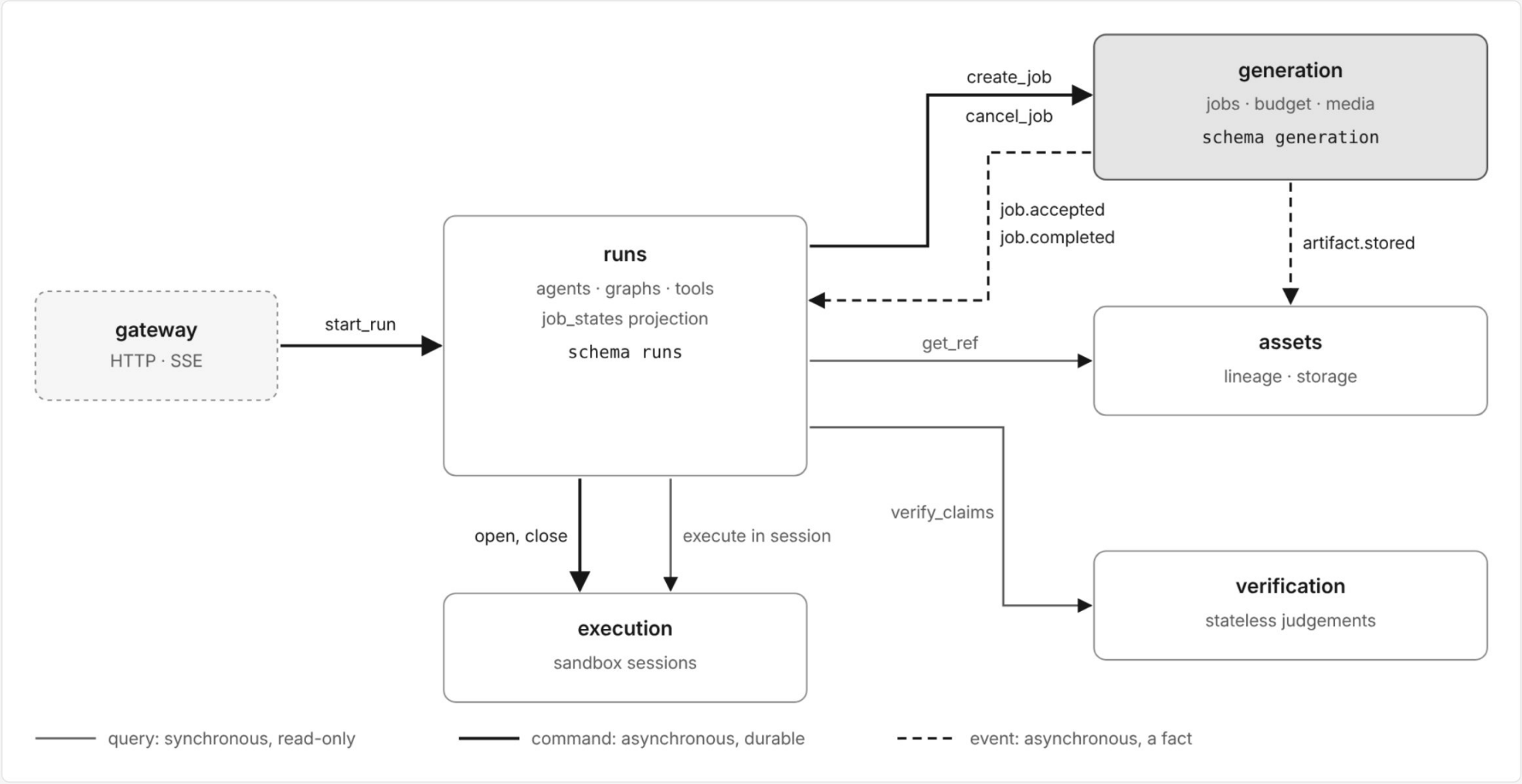
Three kinds of message

08

Pin versions, calibrate thresholds

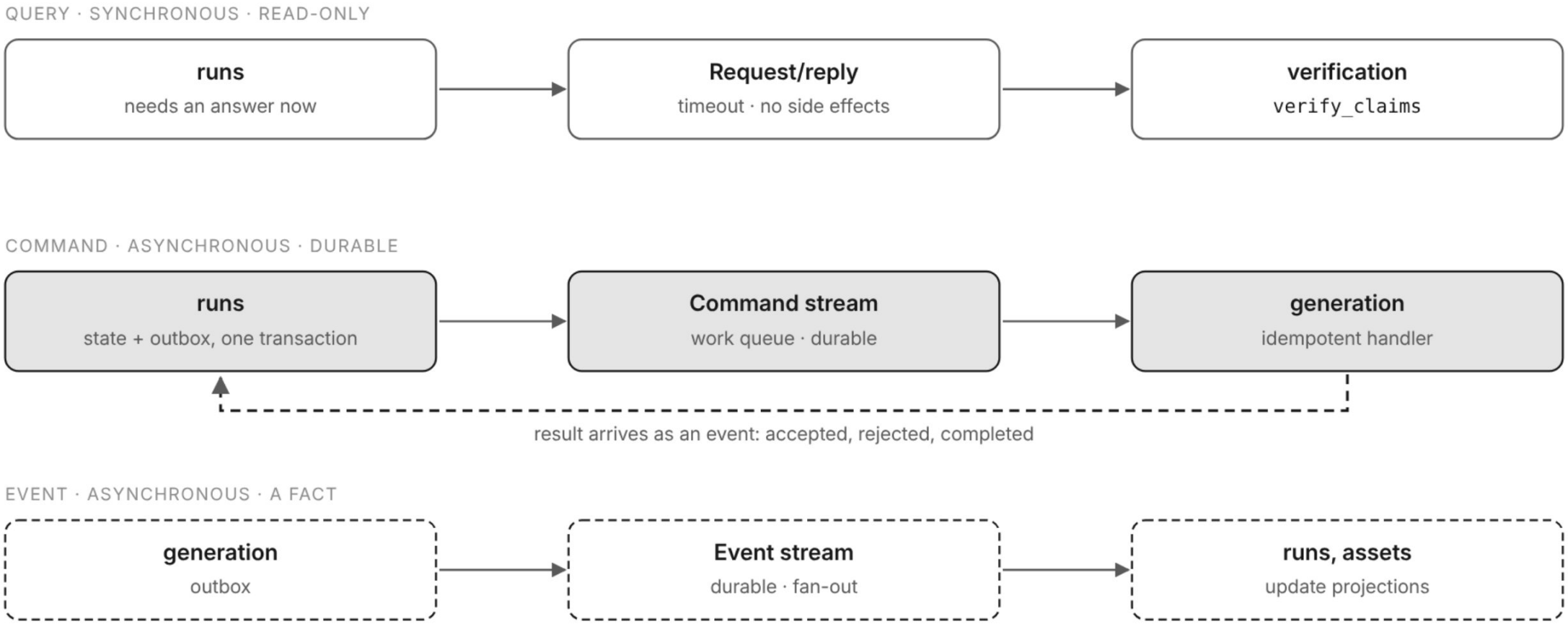
Five modules own their data and meet only through contracts

runs, generation, verification, assets and execution each keep one schema.



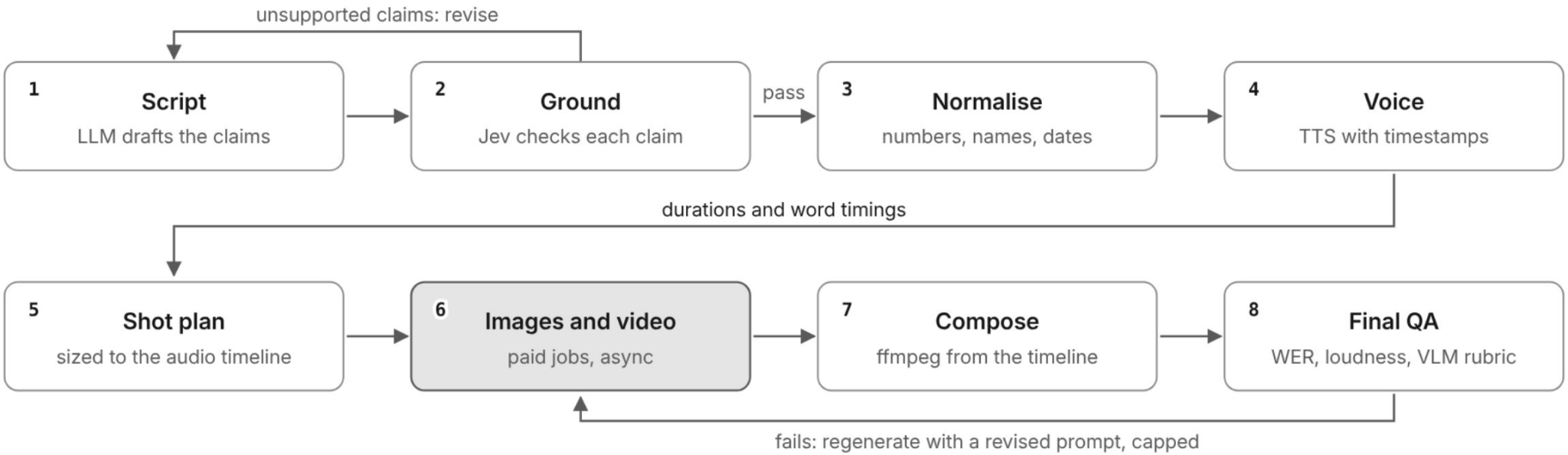
Only three kinds of message cross a module boundary

A query reads, a command asks, an event announces. Commands are durable.



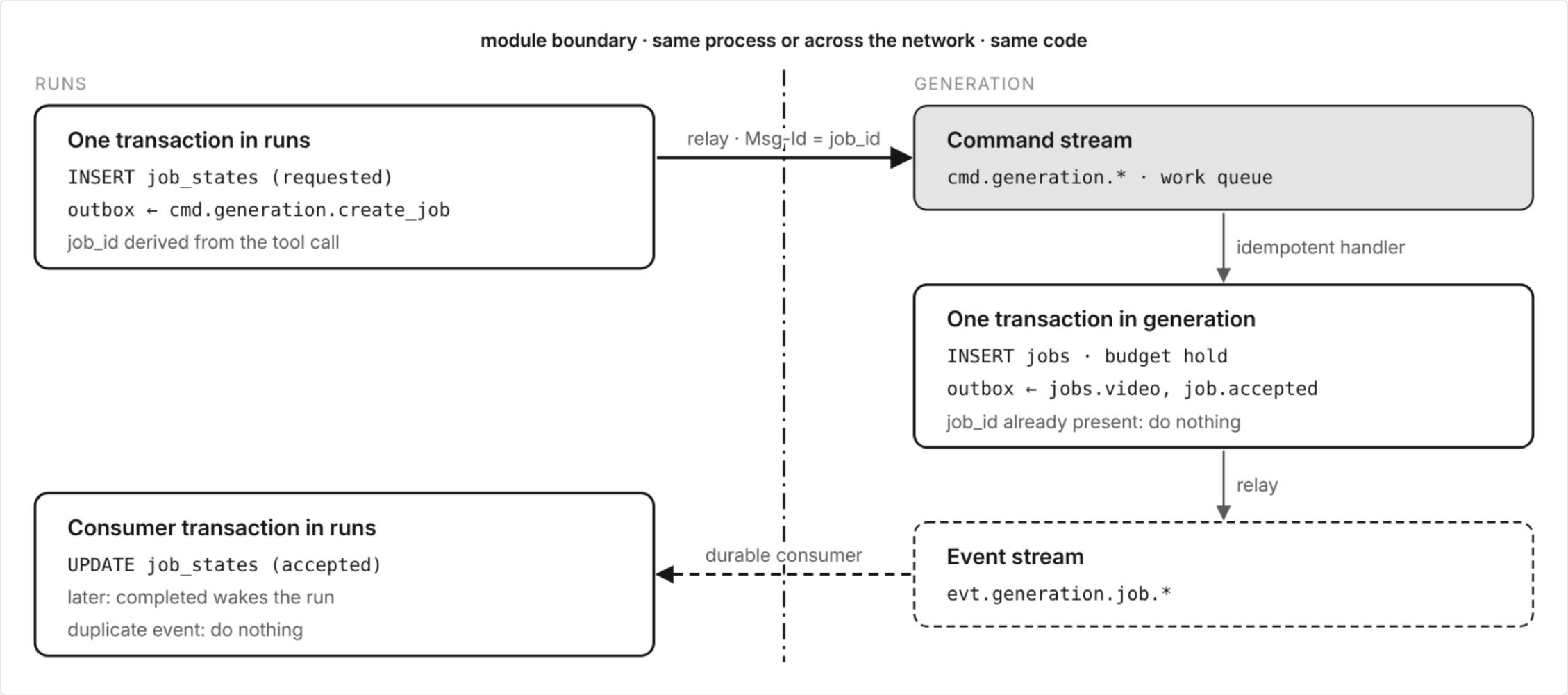
Cheap checks gate every expensive step

Claims are grounded with Jev before voice, and voice fixes the timings before any paid video.



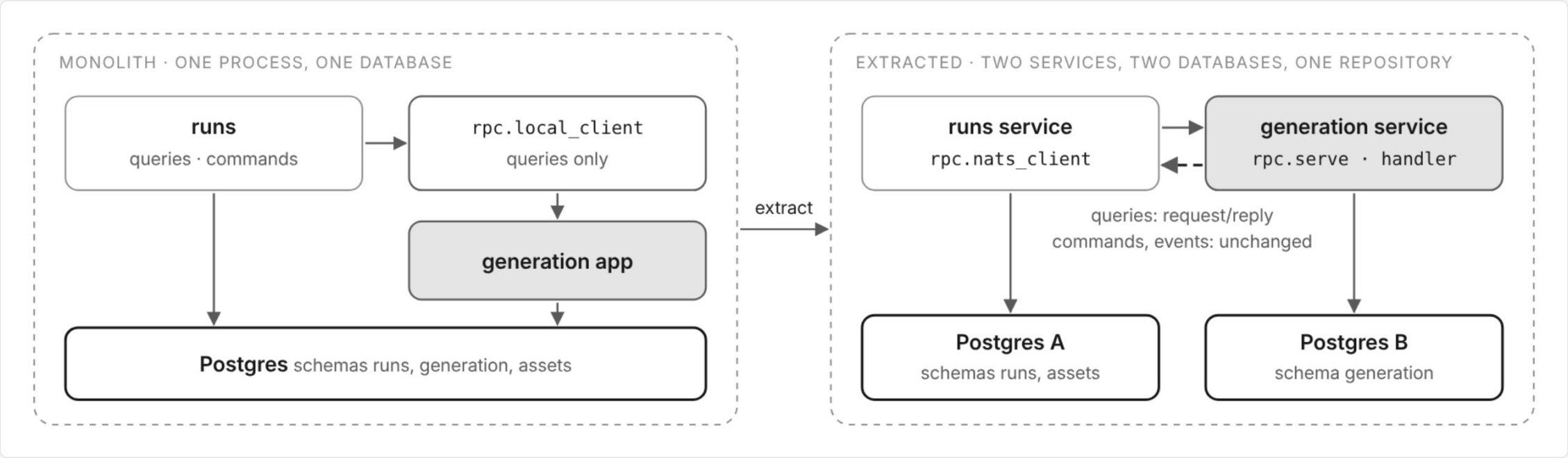
A paid job travels as a durable command, never a call

Each side commits in its own transaction, and a replayed tool call yields the same job_id.



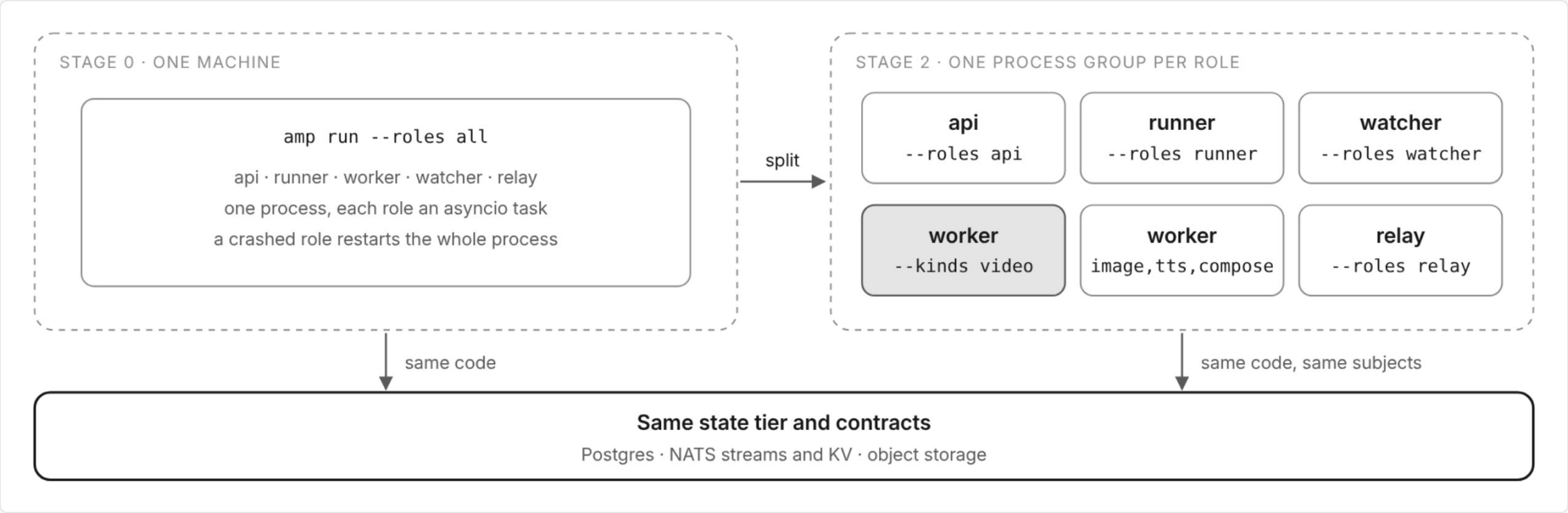
Extracting a module changes only how queries travel

Commands and events already use NATS, so the schema moves and the code stays in one repository.



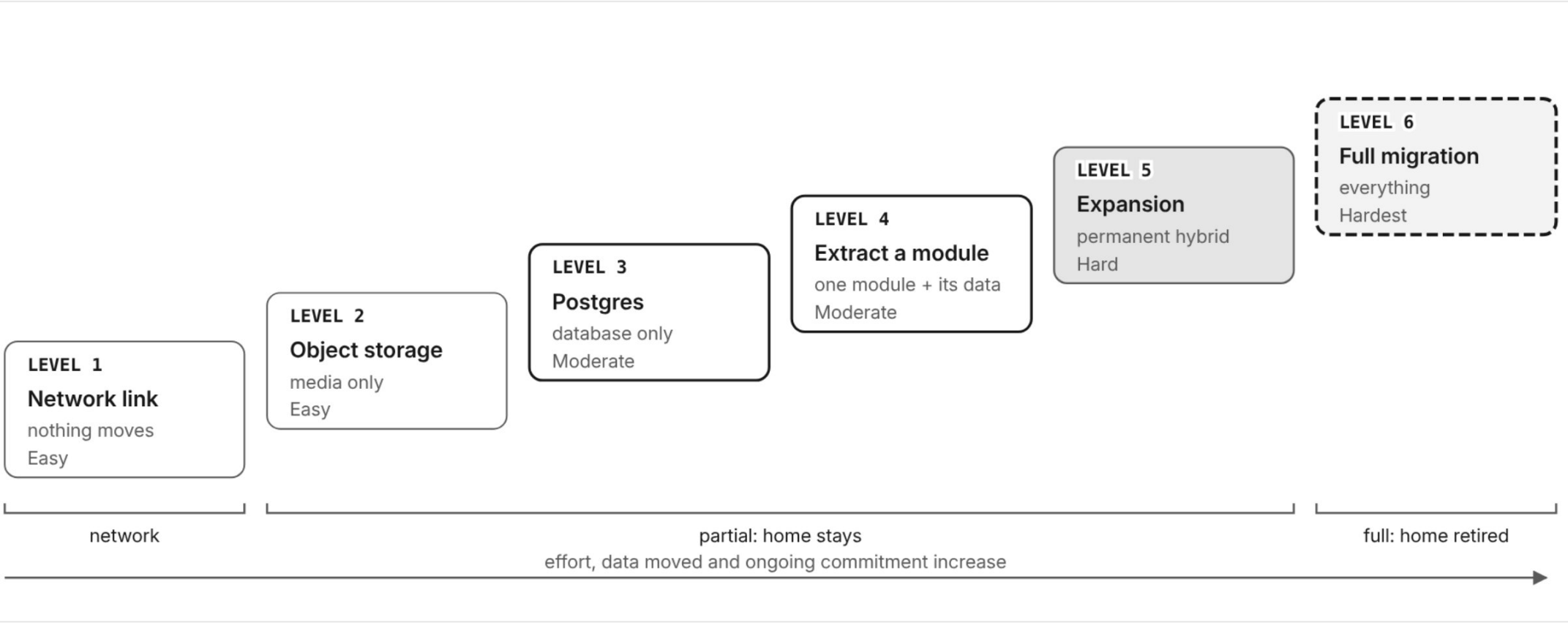
Splitting roles is a deployment change, not a rewrite

Start with --roles all; the video worker is usually the first role worth isolating.



Moving to another cloud is a six-level ladder, easiest first

Level 4, extracting one module with its data, is preferred; a module is never split across providers.



CONCLUSION

Four habits keep an agent platform cheap to change

01

One database is the record

The message bus only delivers, so it can always be rebuilt.

02

Start as a modular monolith

Modules meet only through queries, commands and events, so any can become a service.

03

Gate spend behind cheap checks

Ground claims and fix timings before paying for video, and make every paid call idempotent.

04

Make long work asynchronous

Jobs suspend a run instead of blocking it, and state holds references, not bytes.

If the bus were wiped, nothing would be wrong, only slower.